

Kartikey Pathak

I build machines end to end — mechanical, PCB, firmware, control, and the app on top.

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EDUCATION

Veermata Jijabai Technological Institute (VJTI)

B.Tech, Electrical Engineering — Minor in Robotics (CGPA 8.04)

Mumbai, India

2023 – 2027

Matoshree Prabodhini Jr. College

HSC, Maharashtra State Board — MHT-CET 99.14 percentile

Maharashtra, India

2023

EXPERIENCE

Eyecandy Robotics

Nov 2025 – Present

Electronics / Hardware Engineer

- Design the PCB stack for a legged companion robot: compute boards (Raspberry Pi Zero 2W and Milk-V SoC carriers), 1S/2S power and charging boards (BQ25887, MP2672, TPS562202 buck rails), a servo motor-driver board, a microphone board and inter-board connector boards — schematic through layout, DFM and fabrication output.
- Board-level integration of an MPU6050 IMU, a Feetech servo bus with logic buffering, per-leg force-sensitive-resistor conditioning (LM393), PAM8403 audio and a ReSpeaker mic array; laid out test points and solder-jumper options so each subsystem can be probed or bypassed during bring-up.
- Wrote the fleet-provisioning tooling that clones one built Raspberry Pi runtime image onto many robots with unique machine IDs, hostnames and SSH host keys, so a fleet coexists on one network without IP or config collisions.

NishCorp Technologies

May – Nov 2025

Robotics Control Systems Intern

Hybrid

- Turned the CAD of a stair-climbing wheelchair into a working simulation: CAD to URDF, spawned in Gazebo/RViz, differential-drive kinematics, IMU plugin and teleop — so motion and steering could be studied before anything was built in steel.
- Helped build the half-scale hardware prototype through 3D printing, assembly, motor electronics and field testing; it could sit and stand under its own power.
- The simulation work formed part of the technical case that helped the startup raise Rs 13 lakh in grants.

Krishna Defence and Allied Industries Ltd.

May – Jul 2025

Research Intern, Defence Technology

at IIT Gandhinagar

- Solved indoor localization for an autonomous forklift in a naval dockyard, where GPS is blocked by steel and wheel odometry drifts; surveyed Bluetooth, LoRa, RFID, radar and vision before selecting ultra-wideband ranging.
- Wrote custom double-sided two-way ranging firmware for the Qorvo DWM3001CDK, tuning channel, preamble length and PAC size and re-deriving timing delays per configuration — extending usable range from the stock 45 m to **315 m**.
- Ran SLAM and NAV2 on a Raspberry Pi 4B fusing UWB, IMU and LIDAR; scaled testing from a 2×2 m room to a 50×60 m yard and validated against a surveyor's total station rather than our own code: **0.4% error over 10,000 m²**.

PROJECTS

PCB Axial-Flux BLDC Motor | KiCAD, FEMM, Python, STM32, FOC — final year project

2026 – Present

- Designing a coreless axial-flux motor whose stator is the PCB: 100 mm 2-layer board, 6 wedge coils over 8 poles, dual 3D-printed rotor carrying 16 N42 magnets across a 0.5 mm air gap. Coil geometry is generated programmatically rather than drawn by hand.
- Built the magnetic model myself after establishing that a top-down 2D cut cannot yield torque: unrolled the machine on a cylinder into five radial slices to put coil current out of plane, then extracted Kt, Kv, back-EMF and ripple from the solved air-gap flux. Predicted 40.9 mN·m/A rms, 496 rpm/V and zero cogging torque.
- Simulation caught a design error before fabrication — the plastic rotor gives away $3.3\times$ the torque of the intended steel-backed design — and showed the motor is thermally, not magnetically, limited. Sized 6 mm steel back plates as the fix by checking plate saturation.
- Companion driver firmware on an STM32G431 with a DRV8311H: centre-aligned 3-phase PWM at 20 kHz, a 40 kHz control ISR with zero-sequence injection, AS5600 magnetic encoder over I2C and DMA-fed phase current sensing.

Smart Switch Board | ESP32, ESP-IDF, FreeRTOS, MQTT/TLS, BLE, Flutter, CAD

2025 – Present

- Built an internet-controlled mains wall switch across every layer alone: CAD enclosure, a relay and optocoupler board isolating 220 V from a 3.3 V MCU, the firmware, a TLS MQTT broker and a Flutter app.
- Held to a hard rule that the physical switches keep working with WiFi, cloud and app all down — WiFi provisioning over BLE with nothing hardcoded, state persisted in NVS across power cuts, and a 5-second heartbeat keeping switch, firmware, broker and app agreed on state. Installed and running at home.

- e-Yantra Warehouse Drone, IIT Bombay** | ROS2, OpenCV, ArUco, PID, path planning Oct 2024 – Jan 2025
- Tuned PID control for stable hover and precise waypoint motion, and wrote the vision that detects and decodes ArUco markers for localization and pick-and-place in a simulated warehouse. **National semifinalist.**
- MARIO — 3-DOF Teaching Arm** | ROS2, Gazebo Fortress, micro-ROS, ESP32, URDF, C++ Jul – Aug 2024
- Ported my college robotics society’s teaching arm off end-of-life Gazebo Classic — model, plugins and launch files — rewrote teleop for beginners, and unified the ESP32/micro-ROS pipeline so one command set drives both simulation and hardware. Still runs the ROS2 workshops.
- Also built** | vision/ML, embedded, mechanical 2023 – 2025
- **Coin sorter** — 24-hour hackathon; a camera plus a transfer-learned CNN on a Raspberry Pi classifies Rs 1/2/5/10 and servo gates route each coin, above 95% on Rs 5 and Rs 10.
 - **Voice-controlled pick-and-place arm** — speech recognition to object detection to inverse kinematics in ROS2/Gazebo. **Kurma** — statically stable quadruped with a turtle-derived gait, Mass Robotics entrant.
 - **Self-balancing robot** — complementary-filter sensor fusion and PID on ESP-IDF; the society now teaches on kits of it. **Maze solver** — IR array, 100 Hz loop, left-hand-rule mapping with shortest-path replay, 4th place.
 - **Gesture-controlled car** — glove IMU over ESP-NOW. Plus a cat feeder and a river-cleaning boat.

TEACHING & LEADERSHIP

- Society of Robotics and Automation, VJTI** Aug 2024 – Mar 2025
Active Member — workshops and mentoring VJTI, Mumbai
- Taught workshops on embedded C, ESP32 and sensor fusion (MPU6050, complementary filters) and mentored junior teams through their first self-balancing and line-following robots. The society manufactures and assembles the robot kits these workshops run on; **over 200 students** have learned on them.
- Pratibimb, VJTI Cultural Festival** Aug 2023 – Mar 2025
Department Head, Electrical Engineering VJTI, Mumbai
- Led the Electrical Engineering contingent to its first overall win in the competition’s 26-year history. Separately directed a short film, *Plot Se Panga*, which took first prize across all branches.

TECHNICAL SKILLS

Languages: C, C++, Python, Dart, Lua, TypeScript

Robotics: ROS & ROS2, Gazebo (Classic & Fortress), RViz, URDF, micro-ROS, SLAM, NAV2

Control: PID tuning, sensor fusion, forward & inverse kinematics, gait design, field-oriented control

Embedded: ESP32 (ESP-IDF, FreeRTOS, ESP-NOW), STM32, Raspberry Pi, ARM Cortex-M, I2C/SPI/UART, PWM, ADC/DMA, BLE, MQTT

Electronics: KiCAD schematic & layout, DFM and fabrication output, buck regulators, battery charging, motor drivers, mains isolation, soldering and bring-up with scope and logic analyzer

Simulation & analysis: FEMM magnetostatics, Gazebo physics, OnShape CAD, 3D printing

Vision & ML: OpenCV, ArUco, TensorFlow, CNNs and transfer learning

Tools: Linux, Git, shell scripting, Astro (built my own portfolio site)